

CLINICAL TEST PROTOCOL

Protocol Number: UI02	Title: Usability Validation
Project Number: 1520	Project Name: Intraoral Imager
Usability Specification Number: 4044	PPS Number: SP1-4000

1.0 PURPOSE

To collect data on the usability of software and hardware user interfaces byOrmco-trained and certified operators in a clinical environment.

Five licensed orthodontic clinicians will be trained and certified in system operation by Ormco-approved trainers. Each clinician will then perform tasks based on the user interface requirements for primary operating functions listed in SP1-4044, *Digital Impression System Usability Specification*.

2.0 SPONSOR

CLINICIAN/STAFF MEMBER

Sponsor	Coordinating Sponsor (If applicable)
[Name, Position Title]	[Name, Department]
Ormco Corporation	Ormco Corporation
1717 W. Collins Ave.	[Address]
Orange, CA 92867	[City, State, Zip]

3.0 DEVICE DESCRIPTION

The Digital Impression System for Orthodontic Use is a handheld intraoral 3D imager intended for use inside the human oral cavity to digitally capture the three-dimensional topography of teeth, gingiva, and/or palate in clinical settings. It is intended to replace traditional impressions, which are the current standard for recording dental and orthodontic features of patients undergoing orthodontic treatment.

The system is intended to be used by orthodontists and orthodontic assistants in an orthodontic office environment. The system consists of a handheld wand connected to a computer which is housed in a base unit. The computer contains proprietary software to acquire, process, and store the digital 3D image data. Patient information is entered into the software using a touchscreen monitor connected to the computer. To capture a 3D image of the patient's dental arch

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and/or bite, the operator moves the wand over the surface of the teeth to be scanned. A video camera inside the wand captures images of the teeth surfaces. Algorithms in the software process these images into a 3D image and display the 3D image on the computer monitor in real time. The software also saves the 3D image data and identifying patient information to be used by orthodontic manufacturers to design and manufacture customized orthodontic appliances, (e.g., braces, retainers or aligners).

Indications for Use

The Digital Impression System for Orthodontic Use is an optical impression system intended for use by dental professionals to record the topographical characteristics of teeth, gingiva, and/or palate. The Digital Impression System is intended for use in conjunction with the production of orthodontic appliances, including aligners.

4.0 SUMMARY OF CLINICAL EVALUATION/BACKGROUND

- 4.1 Number of Subjects** – subjects will include five trained operators and five orthodontic patients. This evaluation is to determine the usability of the system by trained operators, which requires creating digital impressions of representative orthodontic patients.

The binomial probability formula was used to determine the number of operators to test that would result in an 85% chance of detecting usability issues that would affect 1/3 of trained operators. The binomial probability formula is $\log(1 - \text{Chance of Detecting}) / \log(1 - \text{Probability of Occurring})$. Testing larger numbers of operators would reveal a higher percentage of issues that would affect a higher percentage of operators.

The number of patients was determined based on a consideration of patient comfort, and in the interest of gaining additional clinical data.

- 4.2 Duration** – the testing described in this protocol should be able to be completed within one day, but can be spread over several days depending on the availability of trained operators, orthodontic patients, orthodontists, and Ormco testing personnel.

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4.3 Follow-up(s) – not required.

4.4 Objective(s) – to confirm the usability of software and hardware user interfaces when used by Ormco-trained and certified operators.

4.5 Endpoint – when five trained operators have completed and been evaluated on all of the user interface tasks based on the user interface requirements for primary operating functions listed in SP1-4044, *Digital Impression System Usability Specification*, and have met the acceptance criteria for all tasks.

4.6 Justification – in the risk analysis for this device, training is used to mitigate risks; therefore, trained operators must be tested. Dental models are useful for training purposes, but do not sufficiently emulate scanning in the human oral cavity. Therefore, testing must be conducted using human subjects.

5.0 CRITERIA FOR OPERATOR SELECTION

- Gender of operators: N/A
- Age of operators: N/A
- Racial and ethnic origin: N/A
- Inclusion criteria: Any licensed orthodontic clinician who has been trained and certified in the operation and maintenance of the Digital Impression System, and is also licensed and certified to take traditional impressions.
- Exclusion criteria: Any person other than a licensed orthodontic clinician who has been trained or certified in the operation and maintenance of the Digital Impression System.

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6.0 CRITERIA FOR PATIENT SELECTION

- Gender of subjects: N/A (Diagnosis, treatment planning, treatment mechanics and orthodontic clinical results do not vary by gender.)
- Age of subjects: 8-14 years old.
- Racial and ethnic origin: N/A (Diagnosis, treatment planning, treatment mechanics and orthodontic clinical results do not vary by racial or ethnic origin of the subjects.)
- Inclusion criteria: Any patient 8-14 years old who would benefit from a course of orthodontic treatment.
- Exclusion criteria: Any patient with anatomical features that would preclude the successful acquisition of a traditional impression.

7.0 RISK/BENEFIT ASSESSMENT

- 7.1 Potential Risk** – This test protocol poses no significant risks to either patients or operators. Low risks would arise from misuse, such as improper cleaning/disinfecting leading to cross contamination, misuse of the scanning wand leading to mechanical failure, or misuse of the scanning wand leading to a laser safety hazard.
- 7.2 Protection Against Risks** – Risks have been mitigated by requiring operator training before using the system, and by affixing a laser safety warning label directly on the system.
- 7.3 Potential Benefits to the Subjects** – Patients would benefit by not having to undergo the traditional impression process.

8.0 REFERENCES

- 8.1** Standards: EN 62366:2008 , MDD 93/42/EEC, Annex X
- 8.2** Specifications:
- 8.2.1 SP1-4044 *Digital Impression System Usability Specification*
- 8.2.2 SP1-4000 *Product Performance Specification*

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9.0 MATERIALS

Part #	Description	Quantity
249-4000	Digital Impression System	1
249-0004	Tip support and locking collar	1 per patient, sterilized
249-0001	Disposable tip	As required
N/A	Cavicide Wipes	As required

10.0 EQUIPMENT

10.1 Video camera to record usability tests.

11.0 PROCEDURE

- 11.1 Test and validate all user interface requirements for the Primary Operating Functions listed in Section 3.3 of SP1-4044 *Digital Impression System Usability Specification* using the procedures listed in Attachments 14.1 through 14.13. Document the results of the tests on the attachments. Post the results in the Usability Engineering Folder.
- 11.2 Test each requirement by asking operators to perform the actions described under Test Inputs.
- 11.3 DO NOT COACH OPERATORS DURING TESTING. This is not a training session. Ask the operator to perform each task, and record the results.

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- 11.4** Before beginning testing, tell each operator the following: “This is a test of the usability of our system and the effectiveness of our training program. Do not worry if you are unable to successfully complete a task – that simply means that we need to improve our design or our training program. We welcome any and all comments you have regarding using our system – if you find something difficult to use, chances are other users will, too. Thank you for helping us evaluate the usability of our system.”
- 11.5** Record the outcome of each test under Conformance to Specification. Check the Yes box if the Actual Result of the test input is the same as the Expected Result. Check the No box if the Actual Result does not match the Expected Result. If the Actual Result is different than the Expected Result, note the difference in the Actual Result box.
- 11.6** Sign and date each record. Also record the system serial number and software release (version) number.

12.0 REVISION HISTORY

Revision	Date	Description Of Change	Author
1.0	6-26-12	Draft version	D. Hoyle

13.0 APPROVALS

Originator	DPI Engineering	DPI Research & Development	DPI Quality Engineering	DPI Program Management

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14.0 ATTACHMENTS

14.1 Validation Test Record – Move System and Power-On

Setup	• System is powered-off and is in another room or operatory. • Wand is fully assembled with tip support and disposable tip, and is in the wand holder. • Tell the operator: “Please move the system into this operatory and power it on.” The operator is expected to be able to move the system without assistance. • Monitor and record each task listed below. Actual Results only need to be noted if they are different than the Expected Results. • Tasks should be performed in the sequence listed below. A task performed out-of-sequence should be recorded as not conforming to the specification, and noted under Actual Results.				
Usability Specification Number	Task	Expected Results	Actual Results (if different than expected results)	Conformance to Specification	
				Yes	No
UI15	Move the system into the operatory.	Operator can easily move the system without assistance. Operator knows how to disconnect and reconnect the power cable.			

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UI03	Power-on the system.	Operator can power the system on by pressing the button at the lower right corner of the touchscreen. If that doesn't work, the Operator knows to check that the power switch next to the power outlet on the back of the system is in the "On" position.			
Tester and Operator Information	Ormco Tester:	Operator:			
	Print Name: _____ Signature: _____ Date: _____	Print Name: _____ _____ _____ _____ _____			
System Information	System Serial Number: _____	Software Version: _____			

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Post-Test Questions for Operator

1. How easy was it to move the system?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

2. How easy was it to power-on the system?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

3. Do you have any complaints about moving the system?

4. Do you have any complaints about powering-on the system?

5. Do you have any suggestions for improving the system?

6. Is there anything else you would like us to know?

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14.2 Validation Test Record – Cleaning and Disinfecting System Components, Replacing the Tip Support and Disposable Tip

Setup	• System can be powered on or off, but if powered on, operator should log off to prevent inadvertently activating touchscreen commands. • Wand is fully assembled with tip support and disposable tip, and is in the wand holder. • Tell the operator: “Assume that we have just finished capturing a digital impression from a patient. Please prepare the system for the next patient.” • Monitor and record each task listed below. Actual Results only need to be noted if they are different than the Expected Results. • Tasks should be performed in the sequence listed below. A task performed out-of-sequence should be recorded as not conforming to the specification, and noted under Actual Results.				
Usability Specification Number	Task	Expected Results	Actual Results (if different than expected results)	Conformance to Specification	
				Yes	No
UI01	Put on a new pair of powderless examination gloves.	Gloves should be worn for this procedure.			
UI08	Remove wand from wand holder.	Operator easily removes wand from wand holder.			
UI01	Wipe the surfaces of the wand body, wand cable, base unit, wand holder, and touchscreen with a CaviWipe to clean the surfaces of all soil. After cleaning each surface, discard the wipe.	All surfaces should be clean and free of visible contaminants.			

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UI01	Use a second CaviWipe to thoroughly wipe the wand body, wand cable, base unit, wand holder, and touchscreen. Repeated use of the second CaviWipe may be required to ensure that the surface remains visibly wet for 3 minutes at room temperature (69°F/20°C).	Surfaces should remain visibly wet for 3 minutes at room temperature (69°F/20°C).			
UI01	Air-dry the wand body, wand cable, and wand holder. Dry the touchscreen with a clean disposable cloth.	Surfaces should be dry.			
UI01	Remove and discard the used disposable tip.	Discard disposable tip.			
UI02	Disassemble the locking collar and tip support from the wand body, and replace them with a sterilized locking collar and tip support.	Sterilized tip support should be secured properly onto the wand body with the locking collar. (Tester should inspect)			
UI02	Install a new disposable tip.	New disposable tip properly attached to wand. (Tester should inspect)			
UI09	Replace wand in wand holder.	Operator securely seats wand in wand holder.			

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Tester and Operator Information	Ormco Tester:	Operator:			
	Print Name:	Print Name:			
	_____	_____			
	Signature:				

	Date: _____				
System Information	System Serial Number:	Software Version:			
	_____	_____			

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Post-Test Questions for Operator

1. How easy was it to clean the system components?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

2. How easy was it to replace the disposable tip?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

3. How easy was it to replace the tip support and locking collar?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

4. How easy was it to disinfect the system components?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

5. How easy was it to remove the wand from the wand holder?

Very Easy 1 2 3 4 5 Very Difficult

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Comments:

6. How easy was it to put the wand back in the wand holder?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

7. How often do you need to change the disposable tip?

8. Is it okay to disinfect and reuse a disposable tip?

9. When do you need to change the tip support?

10. Is it acceptable to disinfect the tip support with a Cavicide wipe, rather than steam-sterilize it?

11. Do you have any complaints about the cleaning and disinfecting process?

12. Do you have any complaints about replacing the tip support or the disposable tip?

13. Do you have any suggestions for improving the system, or is there anything else you would like us to know?

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14.3 Create a New Case, Enter Patient Information, and Run System Check

Setup	- System has been powered off, then powered back on (System Check only appears prior to the first scan after powering-on). - Touch screen is folded down against the base unit. - Wand is fully assembled with tip support and disposable tip, and is in the wand holder. - Tell the operator: "First, raise the touchscreen and adjust it so that you can see it easily. Next, log in to the system, set up a new case, and then proceed to the Impressions screen." - Monitor and record each task listed below. Actual Results only need to be noted if they are different than the Expected Results. - Tasks should be performed in the sequence listed below. A task performed out-of-sequence should be recorded as not conforming to the specification, and noted under Actual Results.				
Usability Specification Number	Task	Expected Results	Actual Results (if different than expected results)	Conformance to Specification	
				Yes	No
UI06	Raise and adjust the touchscreen.	Touchscreen position is easily adjustable.			
UI07 UI12	Log in and enter all required information on the Case Setup screen	It is easy for the operator to select commands on the touchscreen and enter patient information.			
UI07 UI12	Operator finishes entering all required patient information and clicks Impressions at the bottom of the Case Setup page.	The System Check Screen appears.			

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UI04	Operator understands that they must run a System Check before they can scan. Operator clicks Run System Check.	Operator clicks Run System Check.			
UI04	When System Check is complete, operator clicks Continue on the confirmation screen.	Operator clicks Continue on the confirmation screen to proceed to the Impressions screen.			
Tester and Operator Information	Ormco Tester:	Operator:			
	Print Name:	Print Name:			
	Signature:				
	Date: _____				
System Information	System Serial Number:	Software Version:			

Post-Test Questions for Operator

1. How easy was it to adjust the touchscreen position?

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Very Easy 1 2 3 4 5 Very Difficult

Comments:

2. How easy was it to select commands on the touchscreen?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

3. How easy was it to enter patient information?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

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4. How easy was it to run the System Check?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

5. How often do you need to run a System Check?

6. Do you have any comments or complaints about adjusting or using the touchscreen?

7. Do you have any comments or complaints about entering patient information?

8. Do you have any comments regarding the System Check?

9. Do you have any suggestions for improving the system, or is there anything else you would like us to know?

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14.4 Scan a Lower Arch

Setup	- Operator has logged in, entered patient information, and proceeded to the Impressions screen. - All system components have been cleaned and disinfected. - Wand is fully assembled with a sterilized tip support and a new disposable tip, and is in the wand holder. - Patient is in the operatory chair. - Tell the operator: "Please capture an impression of the lower arch." - Monitor and record each task listed below. Actual Results only need to be noted if they are different than the Expected Results. - Tasks should be performed in the sequence listed below. A task performed out-of-sequence should be recorded as not conforming to the specification, and noted under Actual Results. - Acceptance of a complete digital impression requires inspection and approval by a licensed orthodontist, but trained operators should be able to recognize defects and, if necessary, repeat views. - Defects should not be pointed out to the operator by the tester. - Failure to recognize defects by an operator should be noted under Actual Results.				
Usability Specification Number	Task	Expected Results	Actual Results (if different than expected results)	Conformance to Specification	
				Yes	No
	Operator preparation. (Puts on new examination gloves, places OptraGate in patient's mouth, etc.)	Operator is ready to scan patient. Patient is ready to be scanned.			

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UI10 UI11 UI13	Operator selects Lower Arch on the Impressions screen. Operator places wand in patient's mouth and scans the patient's lower arch.	Operator successfully completes all lower arch views in the proper sequence. Operator visually inspects each view before proceeding to the next view. If defects are observed in a view, Operator recognizes the defect and repeats the view. Operator then finishes the lower arch impression, and returns to the Impressions screen.			
Tester and Operator Information	Ormco Tester: Print Name: _____ Signature: _____ Date: _____	Operator: Print Name: _____ Signature: _____			
System Information	System Serial Number: _____	Software Version: _____			

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Post-Test Questions for Operator

1. Did the wand seem to fit comfortably in the patient's mouth?

Very Comfortable 1 2 3 4 5 Very Uncomfortable

Comments:

2. Did the wand seem lightweight?

Lightweight 1 2 3 4 5 Too Heavy

Comments:

3. Was it easy to move the wand in the patient's mouth?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

4. Was it easy to tell how to move the wand based on the scan path images?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

5. How easy was it to collect all of the views needed for this impression?

Very Easy 1 2 3 4 5 Very Difficult

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Comments:

6. Do you have any comments or complaints about the wand size, shape, or weight?
7. Do you have any comments or complaints about moving the wand in the patient's mouth?
8. Do you have any comments regarding the scanning process?
9. Do you have any suggestions for improving the scanning process?
10. Is there anything else you would like us to know?

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14.5 Scan Upper Arch

Setup	- Operator has logged in, entered patient information, and proceeded to the Impressions screen. - All system components have been cleaned and disinfected. - Wand is fully assembled with a sterilized tip support and a new disposable tip, and is in the wand holder. - Patient is in the operatory chair. - Tell the operator: "Please capture an impression of the upper arch." - Monitor and record each task listed below. Actual Results only need to be noted if they are different than the Expected Results. - Tasks should be performed in the sequence listed below. A task performed out-of-sequence should be recorded as not conforming to the specification, and noted under Actual Results. - Acceptance of a complete digital impression requires inspection and approval by a licensed orthodontist, but trained operators should be able to recognize defects and, if necessary, repeat views. - Defects should not be pointed out to the operator by the tester. - Failure to recognize defects by an operator should be noted under Actual Results.				
Usability Specification Number	Task	Expected Results	Actual Results (if different than expected results)	Conformance to Specification	
				Yes	No
	Operator preparation. (If not already done for a previous scan)	Operator is ready to scan patient. Patient is ready to be scanned.			

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UI10 UI11 UI14	Operator selects Upper Arch on the Impressions screen. Operator places wand in patient's mouth and scans the patient's upper arch.	Operator successfully completes all upper arch views in the proper sequence. Operator visually inspects each view before proceeding to the next view. If defects are observed in a view, Operator recognizes the defect and repeats the view. Operator then finishes the upper arch impression, and returns to the Impressions screen.			
Tester and Operator Information	Ormco Tester: Print Name: _____ Signature: _____ Date: _____	Operator: Print Name: _____ Signature: _____			
System Information	System Serial Number: _____	Software Version: _____			

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Post-Test Questions for Operator

1. Did the wand seem to fit comfortably in the patient's mouth?

Very Comfortable 1 2 3 4 5 Very Uncomfortable

Comments:

2. Was it easy to move the wand in the patient's mouth?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

3. Was it easy to tell how to move the wand based on the scan path images?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

4. How easy was it to collect all of the views needed for this impression?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

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5. Do you have any comments or complaints about moving the wand in the patient's mouth?
6. Do you have any comments regarding the scanning process?
7. Do you have any suggestions for improving the scanning process?
8. Is there anything else you would like us to know?

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14.6 Run a Bite Scan

Setup	- Operator has logged in, entered patient information, and proceeded to the Impressions screen. - All system components have been cleaned and disinfected. - Wand is fully assembled with a sterilized tip support and a new disposable tip, and is in the wand holder. - Patient is in the operatory chair. - Tell the operator: "Please capture a bite impression." - Monitor and record each task listed below. Actual Results only need to be noted if they are different than the Expected Results. - Tasks should be performed in the sequence listed below. A task performed out-of-sequence should be recorded as not conforming to the specification, and noted under Actual Results. - Acceptance of a complete digital impression requires inspection and approval by a licensed orthodontist, but trained operators should be able to recognize defects and, if necessary, repeat views. - Defects should not be pointed out to the operator by the tester. - Failure to recognize defects by an operator should be noted under Actual Results.				
Usability Specification Number	Task	Expected Results	Actual Results (if different than expected results)	Conformance to Specification	
				Yes	No
	Operator preparation. (If not already done for a previous scan)	Operator is ready to scan patient. Patient is ready to be scanned.			

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UI10 UI11 UI15	Operator selects CR Bite or CO Bite on the Impressions screen. Operator places wand in patient's mouth and scans the patient's upper arch.	Operator successfully completes the left and right bite views in the proper sequence. Operator visually inspects each view before proceeding to the next view. If defects are observed in a view, Operator recognizes the defect and repeats the view. Operator then finishes the bite impression, and returns to the Impressions screen.			
Tester and Operator Information	Ormco Tester: Print Name: _____ Signature: _____ Date: _____	Operator: Print Name: _____ Signature: _____			
System Information	System Serial Number: _____	Software Version: _____			

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Post-Test Questions for Operator

1. Did the wand seem to fit comfortably in the patient's mouth?

Very Comfortable 1 2 3 4 5 Very Uncomfortable

Comments:

2. Was it easy to move the wand in the patient's mouth?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

3. Was it easy to tell how to move the wand based on the scan path images?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

4. How easy was it to collect all of the views needed for this impression?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

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5. What is the difference between a CO Bite and a CR Bite?
6. Do you have any comments or complaints about moving the wand in the patient's mouth?
7. Do you have any comments regarding the scanning process?
8. Do you have any suggestions for improving the scanning process?
9. Is there anything else you would like us to know?

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14.7 Inspect Images

Setup	- Operator has logged in, entered patient information, and proceeded to the Impressions screen. - All system components have been cleaned and disinfected. - Wand is fully assembled with a sterilized tip support and a new disposable tip, and is in the wand holder. - Patient is in the operatory chair. - Operator has completed at least one view. - Tell the operator: "Please show me how you inspect images using the touchscreen." - Monitor and record each task listed below. Actual Results only need to be noted if they are different than the Expected Results. - Tasks should be performed in the sequence listed below. A task performed out-of-sequence should be recorded as not conforming to the specification, and noted under Actual Results. - Acceptance of a complete digital impression requires inspection and approval by a licensed orthodontist, but trained operators should be able to recognize defects and, if necessary, repeat views.				
Usability Specification Number	Task	Expected Results	Actual Results (if different than expected results)	Conformance to Specification	
				Yes	No
UI19	Operator manipulates image by touching and dragging on the touchscreen (to rotate image). Operator presses Image Tools and uses the Image Tools dialog to inspect the image	Operator demonstrates that he/she knows how to manipulate and inspect images using the touchscreen and Image Tools.			

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Usability Specification Number: SP1-4044	PPS Number: SP1-4000

Tester and Operator Information	Ormco Tester:	Operator:			
	Print Name:	Print Name:			
	_____	_____			
	Signature:				

	Date: _____				
System Information	System Serial Number:	Software Version:			
	_____	_____			

Post-Test Questions for Operator

1. Is it easy to use the touchscreen to inspect images?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

2. Are the Image Tools easy to use?

Very Easy 1 2 3 4 5 Very Difficult

Comments

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3. Do you have any comments or complaints about inspecting images?
4. Do you have any suggestions for improving this process?

Is there anything else you would like us to know?

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14.8 Redo a Previous Scan

Setup	- Operator has logged in, entered patient information, and proceeded to the Impressions screen. - All system components have been cleaned and disinfected. - Wand is fully assembled with a sterilized tip support and a new disposable tip, and is in the wand holder. - Patient is in the operatory chair. - Operator has completed a view. - Monitor and record each task listed below. Actual Results only need to be noted if they are different than the Expected Results. - Tasks should be performed in the sequence listed below. A task performed out-of-sequence should be recorded as not conforming to the specification, and noted under Actual Results. - Acceptance of a complete digital impression requires inspection and approval by a licensed orthodontist, but trained operators should be able to recognize defects and, if necessary, repeat views. - Defects should not be pointed out to the operator by the tester. - Failure to recognize defects by an operator should be noted under Actual Results.				
Usability Specification Number	Task	Expected Results	Actual Results (if different than expected results)	Conformance to Specification	
				Yes	No
UI18	Tester asks Operator: "How would you go about skipping this scan?"	Operator indicates that he/she would press the Skip button.			
UI17	Tester tells Operator: "Please redo that last scan."	Operator clicks the Redo Previous Scan button and repeats the previous scan.			

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Tester and Operator Information	Ormco Tester:	Operator:			
	Print Name:	Print Name:			
	_____	_____			
	Signature:				

	Date: _____				
System Information	System Serial Number:	Software Version:			
	_____	_____			

Post-Test Questions for Operator

1. Is it easy to skip a scan?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

2. Was it easy to redo a scan?

Very Easy 1 2 3 4 5 Very Difficult

Comments

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After you clicked Redo Previous Scan, was the confirmation message easy to understand?

Very Easy 1 2 3 4 5 Very Difficult

Comments

3. Do you have any comments or complaints about redoing a scan?
4. Do you have any suggestions for improving this process?
5. Is there anything else you would like us to know?

CLINICAL TEST PROTOCOL

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Usability Specification Number: SP1-4044	PPS Number: SP1-4000

14.9 Run a Fill Scan

Setup	- Operator has logged in, entered patient information, and proceeded to the Impressions screen. - All system components have been cleaned and disinfected. - Wand is fully assembled with a sterilized tip support and a new disposable tip, and is in the wand holder. - Patient is in the operatory chair. - Operator has completed all views of an impression, but has not completed (post-processed) the impression. - Monitor and record each task listed below. Actual Results only need to be noted if they are different than the Expected Results. - Tasks should be performed in the sequence listed below. A task performed out-of-sequence should be recorded as not conforming to the specification, and noted under Actual Results. - Acceptance of a complete digital impression requires inspection and approval by a licensed orthodontist, but trained operators should be able to recognize defects and, if necessary, repeat views. - Defects should not be pointed out to the operator by the tester. - Failure to recognize defects by an operator should be noted under Actual Results.				
Usability Specification Number	Task	Expected Results	Actual Results (if different than expected results)	Conformance to Specification	
				Yes	No
UI16	Tester tells Operator: "Please run a fill scan on any buccal surface."	Operator selects the Fill button, then selects a starting point for the fill scan on the occlusal surface, and then proceeds to run the Fill scan.			

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Tester and Operator Information	Ormco Tester:	Operator:			
	Print Name:	Print Name:			
	_____	_____			
	Signature:				

	Date: _____				
System Information	System Serial Number:	Software Version:			
	_____	_____			

Post-Test Questions for Operator

1. Is it easy to run a fill scan?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

2. Can you run more than one fill scan?

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3. Do you have any comments or complaints about running a Fill scan?
4. Do you have any suggestions for improving this process?
5. Is there anything else you would like us to know?

CLINICAL TEST PROTOCOL

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Usability Specification Number: SP1-4044	PPS Number: SP1-4000

14.10 Complete and Approve a Digital Impression

Setup	- Operator has completed the lower arch, upper arch, and a bite scan, including any fill scans. - Monitor and record each task listed below. Actual Results only need to be noted if they are different than the Expected Results. - Tasks should be performed in the sequence listed below. A task performed out-of-sequence should be recorded as not conforming to the specification, and noted under Actual Results. - Defects should not be pointed out to the operator by the tester. - Acceptance of a complete digital impression requires inspection and approval by a licensed orthodontist. - Five orthodontists will review each digital impression and render their professional opinion on the quality of the digital impression. For comparison purposes, a stone model (made from a traditional impression of the patient) for each patient will be provided. A digital impression will be considered acceptable after review and acceptance by a majority (three or more) of the five orthodontists.				
Usability Specification Number	Task	Expected Results	Actual Results (if different than expected results)	Conformance to Specification	
				Yes	No
UI20	Tester asks Operator: "Is this impression complete, and ready to be inspected and approved?"	Operator indicates that the impression is ready to be inspected and approved.			

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UI20	Orthodontist(s) inspects the digital impression and determines whether or not the digital impression is clinically acceptable. If the impression is deemed clinically acceptable, the orthodontist(s) will sign below. If the impression is deemed clinically acceptable, the orthodontist will click Complete Impressions, and then click Yes on the confirmation pop-up.	Orthodontists inspect the digital impression and deem it clinically acceptable. Five orthodontists will review each digital impression and render their professional opinion on the quality of the digital impression. For comparison purposes, a stone model (made from a traditional impression) for each patient will be provided. A digital impression will be considered acceptable after review and acceptance by a majority (three or more) of the five orthodontists.			
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Tester and Operator Information	Ormco Tester:	Operator:			
	Print Name:	Print Name:			
	_____	_____			
	Signature:				

	Date: _____				
System Information	System Serial Number:	Software Version:			
	_____	_____			
UI20	Orthodontist signature (if clinically acceptable).	Orthodontist #1:			
		Print Name:			

		Signature:			

	Date: _____				

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UI20	Orthodontist signature (if clinically acceptable).	Orthodontist #2: Print Name: _____ Signature: _____ Date: _____			
UI20	Orthodontist signature (if clinically acceptable).	Orthodontist #3: Print Name: _____ Signature: _____ Date: _____			

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UI20	Orthodontist signature (if clinically acceptable).	Orthodontist #4: Print Name: _____ Signature: _____ Date: _____			
UI20	Orthodontist signature (if clinically acceptable).	Orthodontist #5: Print Name: _____ Signature: _____ Date: _____			

Post-Test Questions for Orthodontists

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1. Was it easy to inspect the digital impression?

Orthodontist #1 – Very Easy 1 2 3 4 5 Very Difficult
Orthodontist #2 – Very Easy 1 2 3 4 5 Very Difficult
Orthodontist #3 – Very Easy 1 2 3 4 5 Very Difficult
Orthodontist #4 – Very Easy 1 2 3 4 5 Very Difficult
Orthodontist #5 – Very Easy 1 2 3 4 5 Very Difficult

Comments:

2. Is it easy to use the digital impression to order orthodontic appliances?

Orthodontist #1 – Very Easy 1 2 3 4 5 Very Difficult
Orthodontist #2 – Very Easy 1 2 3 4 5 Very Difficult
Orthodontist #3 – Very Easy 1 2 3 4 5 Very Difficult
Orthodontist #4 – Very Easy 1 2 3 4 5 Very Difficult
Orthodontist #5 – Very Easy 1 2 3 4 5 Very Difficult

Comments

3. Do you have any comments or complaints about inspecting and approving a digital impression?

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4. Do you have any suggestions for improving this process?

5. Is there anything else you would like us to know?

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14.11 Add, Edit, and Remove Operators

Setup	- Operator is logged in as Admin - Monitor and record each task listed below. Actual Results only need to be noted if they are different than the Expected Results.				
Usability Specification Number	Task	Expected Results	Actual Results (if different than expected results)	Conformance to Specification	
				Yes	No
UI21	Tester asks Operator: "Please add a new operator, and also explain the differences between different types of users."	- Operator presses the System icon, then selects Operators. - Operator selects Add Operator on the Add/Modify Operators screen. - Operator enters information for the new operator, and explains the differences between different types of users. - Operator selects Add to add the new operator.			

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UI21	Tester asks Operator: "Please modify an existing operator."	1. Operator presses the System icon, then selects Operators. 2. Operator modifies information by typing in the boxes on the Add/Modify Operators screen. -AND- Operator modifies information by selecting Modify Operator, updating the information on the Modify Operator screen, and then clicking Modify to save changes. Operator must demonstrate that he/she is aware of both methods for modifying an operator.			
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CLINICAL TEST PROTOCOL

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UI21	Tester asks Operator: "Please delete the operator you just created."	1. Operator presses the System icon, then selects Operators. 2. Operator selects the trash can icon for the user on the Add/Modify Operators screen.			
Tester and Operator Information	Ormco Tester: Print Name: _____ Signature: _____ Date: _____	Operator: Print Name: _____ _____			
	System Information	System Serial Number: _____	Software Version: _____		

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Post-Test Questions for Operator

1. Was it easy to add an operator?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

2. Was it easy to edit an operator?

Very Easy 1 2 3 4 5 Very Difficult

Comments

3. Was it easy to delete an operator?

Very Easy 1 2 3 4 5 Very Difficult

Comments

4. Was the Add/Modify Operators screen easy to understand?

Very Easy 1 2 3 4 5 Very Difficult

Comments

5. Do you have any comments or complaints about adding, editing, or deleting operators?

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6. Do you have any suggestions for improving this process?

7. Is there anything else you would like us to know?

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14.12 Add or Edit Practice Information

Setup	- Operator is logged in as Admin - Monitor and record each task listed below. Actual Results only need to be noted if they are different than the Expected Results.				
Usability Specification Number	Task	Expected Results	Actual Results (if different than expected results)	Conformance to Specification	
				Yes	No
UI22	Tester asks Operator: "Please show me how you would add or edit information for your practice."	- Operator presses the System icon, then selects Practice. - Operator adds or edits information on the Practice Setup screen, then selects Save to save the changes.			

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Tester and Operator Information	Ormco Tester:	Operator:			
	Print Name:	Print Name:			
	_____	_____			
	Signature:				

	Date: _____				
System Information	System Serial Number:	Software Version:			
	_____	_____			

Post-Test Questions for Operator

1. Was it easy to edit practice information?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

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2. Do you have any comments or complaints about adding or editing practice information?
3. Do you have any suggestions for improving this process?
4. Is there anything else you would like us to know?

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14.13 Validation Test Record – Laser Safety Labeling

Setup	• System can be powered-off or powered-on. • Wand is fully assembled with tip support and disposable tip, and is in the wand holder. • Monitor and record the answer to each question listed below. Actual Results only need to be noted if they are different than the Expected Results. • Tasks should be performed in the sequence listed below. A task performed out-of-sequence should be recorded as not conforming to the specification, and noted under Actual Results.				
Usability Specification Number	Task	Expected Results	Actual Results (if different than expected results)	Conformance to Specification	
				Yes	No
UI23	Tester asks the operator: “Are there any safety hazards associated with the laser beam emitted by the scanning wand?”	Operator answers “yes.”			
UI23	Tester asks the operator: “What are two things you should never do with the laser beam emitted by the scanning wand?”	Operator answers: 1) Never stare into the laser beam. 2) Never view the laser beam directly with an optical instrument.			

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UI23	Tester asks the operator: “Where can you find laser safety information for this system?”	Operator answers: Laser safety information can be found on a label on the back of the system. Note: Laser safety information is also in the User Guide, but the operator needs to demonstrate (without prompting) that he/she is aware of the laser safety label on the back of the system. If he/she is not aware of the laser safety label on the back of the system, it’s a “no” for compliance.			
Tester and Operator Information	Ormco Tester: Print Name: _____ Signature: _____ Date: _____	Operator: Print Name: _____ Signature: _____			

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System Information	System Serial Number:	Software Version:			
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Post-Test Questions for Operator

1. Is it easy to understand the laser safety label on the back of the machine?

Very Easy 1 2 3 4 5 Very Difficult

Comments:

2. Do you have any safety concerns about the laser beam emitted by the wand?
3. Do you have any complaints or suggestions regarding the laser safety label?
4. Do you have any suggestions for improving the system?
5. Is there anything else you would like us to know?